FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2023

TEST DATE: 17-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. $eV_1 = \frac{hC}{\lambda_1} - \phi$

$eV_2 = \frac{hC}{\lambda_2} - \phi$

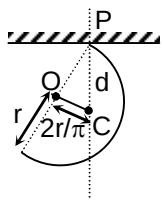
$e(V_2 - V_1) = hc\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)$

$\Rightarrow (V_2 - V_1) = \frac{hc}{e}\left(\frac{\lambda_1 - \lambda_2}{\lambda_1\lambda_2}\right)$

$= \frac{6.6\times10^{-34}\times3\times10^{8}}{1.6\times10^{-19}} \times \frac{100}{6\times10^{-5}} = 2.0625$ volt

2. D

Sol. Use $T = 2\pi\sqrt{\frac{I_P}{mgd}}$



Where $d = \sqrt{r^2 + \left(\frac{2r}{\pi}\right)^2}$ …(i)

$I_0 = mr^2 = I_C + m\left(\frac{2r}{\pi}\right)^2$ …(ii)

and $I_P = I_C + md^2$

$$I_P = I_C + m\left\{r^2 + \left(\frac{2r}{\pi}\right)^2\right\} \quad \ldots(iii)$$

From (ii) and (iii)

$I_P = 2mr^2$

$$T = 2\pi\sqrt{\frac{2mr^2}{mgr\left(1+\frac{4}{\pi^2}\right)^{1/2}}}$$

$$T = 2\pi\sqrt{\frac{2r}{g\left(1+\frac{4}{\pi^2}\right)^{1/2}}}$$

3. B

Sol. $R^3 = Nr^3$

$$\Rightarrow N = \frac{R^3}{r^3}$$

Energy released = $T \times 4\pi\left(Nr^2 - R^2\right)$

$$= 4\pi T\left(\frac{R^3}{r^3}r^2 - R^2\right) = 4\pi TR^2\left(\frac{R-r}{r}\right)$$

So, $\frac{1}{2}mv^2 = 4\pi TR^2\left(\frac{R-r}{r}\right)$

$$\frac{1}{2}\times\frac{4}{3}\pi R^3\rho v^2 = 4\pi TR^2\left(\frac{R-r}{r}\right)$$

$$v = \sqrt{\frac{6T(R-r)}{Rr\rho}}$$

4. A

Sol. Least count = $\frac{0.25}{5\times100}\text{cm} = 5\times10^{-4}$ cm

Reading = 4×0.05 cm + $30 \times 5 \times 10^{-4}$ cm

= 2.15 mm

5. D

Sol. Since, $\mu = \dfrac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$$\cot\left(\frac{A}{2}\right) = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\cos\left(\frac{A}{2}\right) = \sin\left(\frac{A+\delta_m}{2}\right)$$

$$\sin\left(\frac{\pi}{2}-\frac{A}{2}\right)=\sin\left(\frac{A+\delta_m}{2}\right)$$

$\Rightarrow \delta_m = \pi - 2A$

6. A

Sol. $$\frac{mv^2}{r}=\frac{3q^2}{4\pi\varepsilon_0 r^2}\Rightarrow mvr=\frac{3q^2}{4\pi\varepsilon_0 v} \quad \ldots(i)$$

and $$\frac{nh}{2\pi}=mvr \quad \ldots(ii)$$

using (i) and (ii) and putting n = 1

$$\frac{h}{2\pi}=\frac{3q^2}{4\pi\varepsilon_0 v}\Rightarrow v=\frac{3q^2}{2\varepsilon_0 h}$$

7. D

Sol. The bar is at equilibrium. The net force from right or left of a section of BC is 70 KN.

$$\Delta\ell=\frac{F\ell}{AY}=\frac{(70\times10^3)\times1}{1\times2\times10^{11}}$$

$\Delta\ell = 3.5\times10^{-7}$ m

8. C

Sol. T = 20 sec

at t = 2, v = 5 cm/s

$v = A\omega\cos\omega t$

$$v=A\times(2\pi f)\cos\left(\frac{2\pi}{T}t\right)$$

$$5=A\times\left(\frac{2\pi}{20}\right)\cos\left(\frac{2\pi}{20}t\right)$$

$$\Rightarrow \frac{A\pi}{10}\cos36^\circ=5$$

9. A

Sol. The current through circuit $i=\frac{P}{V}=\frac{100\times10^{-3}}{0.5}=0.2$ A

Voltage drop across resistance = 1.5 – 0.5 = 1 V

$$\Rightarrow R=\frac{1}{0.2}=5\Omega$$

10. C

Sol. As t is increasing μ is decreasing, so refraction is taking place from denser to rarer medium as light is proceeding in the medium. Using continuous varying refractive index, let the ray reflect back after travelling $t = t_0$ along the normal, then at this location, angle of refraction is $\frac{\pi}{2}$.

So, $$1\times\sin\theta=\left(\mu_0-kt_0^2\right)\sin\left(\frac{\pi}{2}\right)$$

$$\Rightarrow t_0=\sqrt{\frac{\mu_0-\sin\theta}{k}}$$

11. B

Sol. Compare the given equation with standard equation

$E_y = E_0\cos(\omega t - kx)$

$\therefore\ \omega = 2\pi \times 10^6$

$2\pi\nu = 2\pi \times 10^6$

$\Rightarrow \nu = 10^6$ Hz

$k = \pi \times 10^{-2}$

or $\frac{2\pi}{\lambda} = \pi \times 10^{-2}$

or $\lambda = 200$ m

12. B

Sol. For first diffraction minimum

$a\sin\theta = \lambda$

$\Rightarrow a = \frac{\lambda}{\sin\theta}$

For first secondary maximum, we use

$a\sin\theta' = \frac{3\lambda}{2}$

$\sin\theta' = \frac{3\lambda}{2} \times \frac{1}{a} = \frac{3\lambda}{2} \times \frac{\sin\theta}{\lambda}$

$= \frac{3}{2} \times \sin 30° = \frac{3}{4}$

$\theta' = \sin^{-1}\left(\frac{3}{4}\right)$

13. D

Sol. u = velocity of water through the hole= $\sqrt{2gh}$

R = rate of mass flow = ρAu [A = area of the hole]

F = reaction force on the wall

$= R(u - 0) = \rho Au^2$

For cylinder to just move

$\mu w = \rho Au^2 = \frac{\rho\pi d^2}{4}\left(\sqrt{2gh}\right)^2$

$\therefore\ d = \sqrt{\frac{2\mu w}{\pi\rho hg}}$

14. B

Sol. General equation of travelling wave is

$y = f(ax \pm bt) = f(x \pm vt)$

$\therefore\ y = e^{-(ax^2 + bt^2 + 2\sqrt{ab}xt)}$

$= e^{-(\sqrt{a}x + \sqrt{b}t)^2}$

So, the wave will be travelling along –x axis with speed $\sqrt{\frac{b}{a}}$

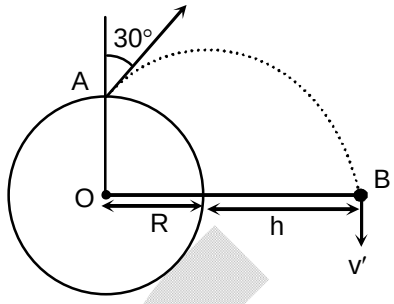
15. B

Sol. By conservation of angular momentum of the body about O

$(mv\sin 30°)R = mv'(R+h)$

$\frac{vR}{2} = \frac{v}{4}(R+h)$

$\Rightarrow h = R$



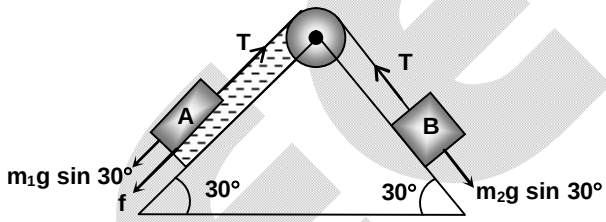
16. A

Sol. $T = m_2 g\sin 30°$ …(i)

$T = m_1 g\sin 30° + \frac{\eta A v}{\ell}$ …(ii)

From (i) and (ii)

$\eta = \frac{(m_2 - m_1)g\ell\sin 30°}{vA} = 3.46\times 10^{-3}$ Pa sec



17. A

Sol. As all the points on the periphery of either ring are at the same distance from point P, the potential at point P due to the whole ring can be calculated as $V = -\frac{GM}{\sqrt{R^2+x^2}}$, where x is the axial distance from the centre of the ring. This expression is independent of the fact whether the distribution of mass is uniform or non-uniform

So, at P, $V = -\frac{GM}{R\sqrt{2}} - \frac{G2M}{R\sqrt{5}} = -\frac{GM}{R}\left[\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}}\right]$

18. A

Sol. $\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$

$1 = \frac{v}{c}$

$v = c$

19. D

Sol. Acceleration when block B is in the liquid

$a_1 = \frac{m_A g - (m_B g - \text{upthrust})}{m_A + 3m}$

$= \frac{m_A g - \left(3mg - \frac{3m}{2\rho}\rho g\right)}{(m_A + 3m)} \uparrow$

Acceleration when block B is outside of the liquid.

$a_2 = \frac{3mg - m_A g}{m_A + 3m}$

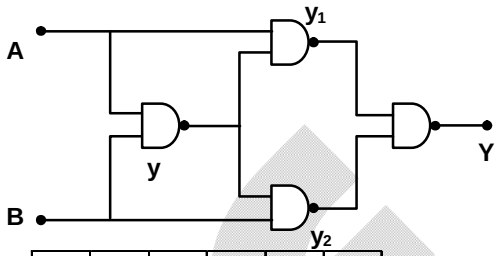
Given $a_1 = a_2$

$\Rightarrow m_A g - \frac{3}{2}mg = 3mg - m_A g$

$$m_A = \frac{9}{4}m$$

20. A

Sol. Output of the gates are marked as y, y_1 and y_2.



A NAND gate gives low output when both the inputs are high, otherwise it gives a high output. Now the truth table.

A	B	y	y_1	y_2	Y
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	0

SECTION – B

21. 680

Sol. $$f' = f\left(\frac{v}{v - v_s\cos\theta}\right)$$

$$f' = 640\left(\frac{340}{340 - \frac{100}{3}\cos\theta}\right)$$

$$f' = 640\left(\frac{340}{340 - \frac{100}{3}\times\frac{3}{5}}\right)$$

$\Rightarrow f' = 680$ Hz

22. 42

Sol. $$F = (\text{Area})\frac{I}{C}$$

Effective area = πR^2

$$F = (\pi R^2)\frac{I}{C}$$

$$F = \frac{22}{7}\times(21\times10^{-2})^2\times\frac{1}{110}\times\frac{1}{3\times10^8}$$

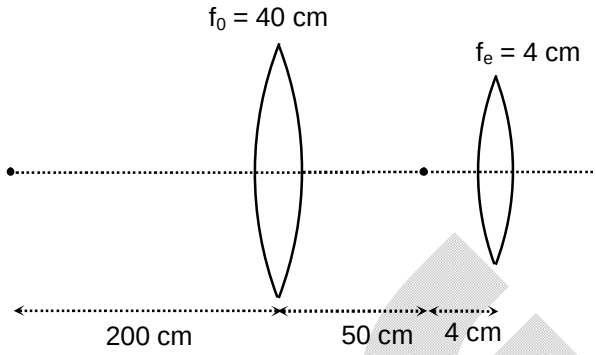
$$F = 42\times10^{-13}\text{N}$$

23. 54

Sol. $\frac{1}{v}-\frac{1}{-200}=\frac{1}{40}$

v = 50 cm

d = 50 + 4 = 54 cm



24. 25

Sol. Here $T_{1/2} = 270$ days

After 540 days, it means two half lives have been completed n = 2

$\frac{N}{N_0}=\left(\frac{1}{2}\right)^n$

$\frac{m}{m_0}=\left(\frac{1}{2}\right)^n$

$\Rightarrow \frac{m}{100}=\frac{1}{4}$

$\Rightarrow$ m = 25 mg

25. 8

Sol. If I is the intensity of the incident unpolarised light, the intensity transmitted by the first is I/2. This is the intensity of incident light on the second polaroid. Intensity transmitted by the second polaroid is $\frac{I}{2}\cos^2\theta$, where θ is the angle between the axes.

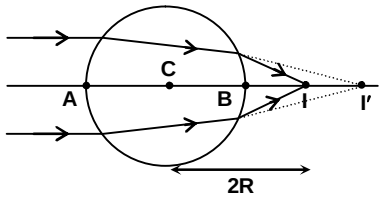
Here sin θ = 3/5 therefore cos θ = 4/5

$\frac{I}{2}\cos^2\theta=\frac{I}{2}\left(\frac{4}{5}\right)^2=\frac{8}{25}I$

$\therefore$ $\frac{8}{25}$ is the required ratio.

26. 3

Sol. from first refaction



$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$

$\frac{n}{v}-\frac{1}{\infty}=\frac{n-1}{R}$

$\frac{n}{v}=\frac{n-1}{R}$

$AI' = v = \frac{nR}{n-1}$

for 2nd refraction

$u = BI' = AI' - 2R$

$=\frac{nR}{n-1}-2R$

$= \frac{(2-n)R}{(n-1)}$

and v = R

For second surface

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\frac{1}{v} - \frac{n}{u} = \frac{1-n}{-R}$

$\frac{1}{R} - \frac{n(n-1)}{(2-n)R} = \frac{n-1}{R}$

$(2-n) - n(n-1) = (n-1)(n-2)$

$n = 4/3$

27. 4

Sol. Let ground state energy (in eV) be E,

Then from given condition,

$E_{2n} - E_1 = 204$ eV

$\frac{E_1}{4n^2} - E_1 = 204$ eV

$E_1\left(\frac{1}{4n^2} - 1\right) = 204$ eV …(i)

and $E_{2n} - E_n = 40.8$ eV

$\frac{E_1}{4n^2} - \frac{E_1}{n^2} = 40.8$ eV

$\Rightarrow E_1 \times -\frac{3}{4n^2} = 40.8$ eV …(ii)

From (i) and (ii)

$\frac{1 - \frac{1}{4n^2}}{\frac{3}{4n^2}} = 5$

n = 2

Now from equation (ii)

$E_1 = -\frac{4}{3}n^2(40.8)$

Or $E_1 = -217.6$ eV

and $E_1 = -13.6\ Z^2$

Therefore

$Z^2 = \frac{E_1}{-13.6} = \frac{-217.6}{-13.6} = 16$

Z = 4

28. 3

Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{3}{25} + \frac{1}{u} = \frac{1}{10}$

$\Rightarrow u = -50$ m

and for final position of the object

$\frac{7}{50}+\frac{1}{u}=\frac{1}{10}$

$\Rightarrow u = -25$ m

Therefore, the shift in object = $50 - 25 = 25$ m

Speed = $\frac{25}{30}$ m/s = 3 km/hr

29. 200

Sol. Fundamental frequency of open pipe $f_0 = \frac{v}{2\ell}$

Third harmonic of the closed pipe $f_C = 3\left(\frac{v}{4}\right)$

Given $3\left(\frac{v}{4\ell}\right)=\frac{v}{2\ell}+100$

Or $\frac{v}{2\ell}=200$ Hz

30. 8

Sol. Shift $\Delta x = \frac{\beta}{\lambda}(\mu-1)t$

Shift due to one plate = $\Delta x_1 = \frac{\beta}{\lambda}(\mu_1-1)$

Shift due to another plate = $\Delta x_2 = \frac{\beta}{\lambda}(\mu_2-1)t$

Net shift $\Delta x = (\Delta x_2 - \Delta x_1)$

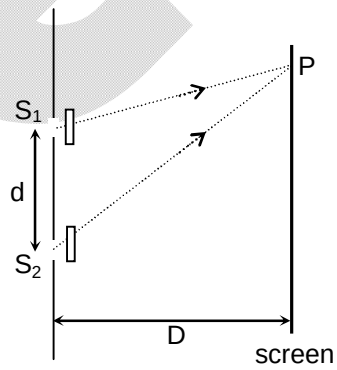
$\Delta x = \frac{\beta}{\lambda}(\mu_2-\mu_1)t$ …(i)

and also $\Delta x = 5\beta$ …(ii)

Hence $5\beta = \frac{\beta}{\lambda}(\mu_2-\mu_1)t$

$\Rightarrow t = \frac{5\lambda}{(\mu_2-\mu_1)} = 8\times10^{-6}$m

t = 8 μm



Chemistry

PART – B

SECTION – A

31. D

Sol. $I_2 + \underset{(\text{hot and conc.})}{NaOH} \longrightarrow NaI + NaIO_3$

32. C

Sol. $6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

33. A

Sol. Because half filled electronic configuration is achieved in Mn^{2+}.

34. D

Sol. Actinoid compounds are more basic than lanthanoid compounds.

35. B

Sol. $[PdCl_4]^{2-}$ has square planar geometry.

36. C

Sol. SO_3 is more harmful than SO_2. SO_2 affects the formation of chlorophyll.

37. D

Sol. $ClO^{\bullet} + O \longrightarrow Cl^{\bullet} + O_2$ reaction is involved in ozone layer depletion.

38. C

Sol. Ag_2CrO_4 is red in colour.

39. D

Sol. Dilute boric acid solution is weak antiseptic for eyes.

40. B

Sol. Bactericidal – Penicillin, Aminoglycosides and ofloxacin.
Bacteriostatic – Tetracyclin.

41. C

Sol. Multimolecular : Sol of sulphur
Macromolecular : Sol of protein, Sol of starch.

42. B

Sol. Moles of acetic acid adsorbed $= (0.6 \times 100 - 0.4 \times 100) \times 10^{-3}$

$= 20 \times 10^{-3}$

Mass of acetic acid adsorbed (W) = $20 \times 10^{-3} \times 60$ g

Mass of acetic acid adsorbed per gram activated charcoal $= \frac{1.2}{4}$

= 0.3 g.

43. C

Sol. $KE = \frac{3}{2}RT \qquad KE \propto T$

$u_{rms} = \sqrt{\frac{3RT}{M}}$ $\quad u_{rms} \propto \sqrt{T}$

$u_{rms} \propto \frac{1}{\sqrt{M}}$

44. D

Sol. $+5.8 = C \times 5.8 + \frac{200 \times 4.2 \times 5.8}{1000}$

$1 = C + 0.84$

$C = 0.16$ kJ/K

45. B

Sol. Calamine : $ZnCO_3$ Argentite : Ag_2S

Willemite : Zn_2SiO_4 Diaspore : $Al_2O_3.H_2O$

46. C

Sol. Extraction of aluminium from concentrated solution is not done by coke.

47. D

Sol. $PE = \frac{\frac{4}{3}\pi r_+^3 + \frac{4}{3}\pi r_-^3}{(2r_-)^3} \times 100$

$= \frac{4 \times \pi}{3 \times 8} \times 100 \left[\left(\frac{r_+}{r_-} \right)^3 + 1 \right]$

$= \frac{\pi}{6} \times 100 \left[(0.732)^3 + 1 \right]$

$= \frac{\pi}{6} \times 100 \times 1.4$

$= 72.8$

≈ 73

48. D

Sol. Due to decrease in the interionic forces.

49. B

Sol. $(n \times nf)_{K_2Cr_2O_7} = (n \times nf)_{CuS}$

$n \times 6 = 2.4 \times 8$

$n = \frac{2.4 \times 8}{6} = 3.2$

50. D

Sol. $n \times 2 + 2 \times 3 + 4 \times 6 - 18 \times 2 = 0$

$n = 3$

SECTION – B

51. 5

Sol. $P_4 + 10Cl_2 \longrightarrow 4PCl_5$

$PCl_5 + 4H_2O(\text{excess}) \longrightarrow H_3PO_4 + 5HCl$

52. 3

Sol. $\Delta T_f = iK_f, m$

$$0.52 = i \times 1.86 \times \frac{0.2}{1.86}$$

$$i = 2.6$$

$$\alpha = \frac{i-1}{n-1}$$

$$(n-1) = \frac{i-1}{\alpha} = \frac{2.6-1}{0.8} = \frac{1.6}{0.8} = 2$$

$$n - 1 = 2$$

$$n = 3$$

53. 462

Sol. $H_2\,(4\text{ atm}) \longrightarrow 2H^+\,(C_1) + 2e^-$

$2H^+\,(C_2) + 2e^- \longrightarrow H_2\,(1\text{ atm})$

$H_2\,(4\text{ atm}) + 2H^+\,(C_2) \longrightarrow 2H^+\,(C_1) + H_2$

For acetic acid

$$\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5}}{10^{-1}}} = 10^{-2}$$

$$\left[H^+\right](C_1) = C\alpha = 10^{-1} \times 10^{-2} = 10^{-3}$$

From NaOH $\left[H^+\right](C_2) = 10^{-11}$

$$E = \frac{0.06}{2} \log \frac{C_2^2 \times 4}{C_1^2 \times 1}$$

$$= 0.03 \log \left(\frac{10^{-22} \times 4}{10^{-6} \times 1} \right)$$

$$= 0.03\left(\log 4 \times 10^{-16}\right)$$

$$= 0.03\left[-16 + 0.6\right]$$

$$= -0.03 \times 15.4$$

$$= -0.462$$

54. 25

Sol. $$d = \frac{Z \times M}{a^3 \times N}$$

$$N = \frac{Z \times M}{d \times a^3} = \frac{2 \times 512}{8 \times \left(8 \times 10^{-8}\right)^3} = \frac{2 \times 512 \times 10^{24}}{8 \times (8)^3}$$

$$= 0.25 \times 10^{24} = 25 \times 10^{22}$$

55. 3

Sol. $2KI + H_2O + O_3 \longrightarrow 2KOH + I_2 + \underset{\text{'Y''}}{O_2}$

Number of unpaired electrons in O_2 = 2

$\mu = \sqrt{n(n+2)}$ B.M.

$= \sqrt{2(2+2)}$ B.M.

$= \sqrt{8}$ B.M.

= 2.83

≈ 3

56. 1

Sol. $60 \times x = 15 \times 0.02 \times 2$

$x = 0.01$ M

57. 414

Sol. $-2573 = [2 \times 814 + 4 \times H_{C-H}] + [5 \times 499] - [8 \times 724 + 4 \times 640]$

$-2573 = [1628 + 4 \times H_{C-H} + 2495] - [5792 + 2560]$

$H_{C-H} = 414$

58. 6

Sol. $NaCl + K_2Cr_2O_7 + H_2SO_4 \longrightarrow \underset{(\text{Orange red vapours})}{CrO_2Cl_2} + K_2SO_4 + H_2O$

$CrO_2Cl_2 + NaOH \longrightarrow \underset{(\text{Yellow solution})}{Na_2CrO_4} + NaCl$

$Na_2CrO_4 + Pb(CH_3COOH)_2 \longrightarrow \underset{(\text{Yellow ppt.})}{PbCrO_4} + CH_3COONa$

59. 21

Sol. $MnO_4^- + Mn^{2+} \longrightarrow MnO_2 + H_2O$

$n_f(MnO_2) = \frac{6}{5},\ n_f(MnO_4^-) = 3$

$(n_f \times n)_{MnO_2} = (n_f \times n)_{KMnO_4}$

$\frac{6}{5} \times n = 3 \times 4.2$

$n = 2.1 \times 5$

$n = 10.5$

$2Y = 2 \times 10.5 = 21$.

60. 6

Sol. $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + CaCl_2 + H_2O$

$n_{NH_4Cl} = \frac{10.7}{53.5} = 0.2$

$n_{Ca(OH)_2} = \frac{M \times V}{1000} = \frac{5 \times 10}{1000} = 0.05$

Limiting reagent is $Ca(OH)_2$

$n_{NH_3} = 0.05 \times 2 = 0.1$

PV = nRT

$0.82 \times V = 0.1 \times 0.082 \times 600$

V = 6 L

Mathematics

PART – C

SECTION – A

61. A

Sol. $z = 1, f(1) = a + ib$

$\Rightarrow a + ib$ must fall on the line $Re(z) = a = \frac{1}{2}$ (By equidistant rule)

$\Rightarrow |a + ib| = 8 \Rightarrow b^2 = \frac{255}{4}$

62. B

Sol. Conceptual

63. A

Sol. If p is a prime such that $100 > p > \frac{200}{3}$, then p and 2p but not 3p belong to [1, 2, …, 200].

While just p belongs to (1, 2, …, 100), so p does not divide ${}^{200}C_{100}$ but consider 61 the largest prime $< \frac{200}{3}$

61^3 divides 200! but only 61 divides 100!

Hence 61 divides ${}^{200}C_{100}$.

64. D

Sol. If $n^2 - 19n + 99 = x^2$

$\Rightarrow n^2 - 19n + 99 - x^2 = 0$

$n = \frac{19 \pm \sqrt{4x^2 - 35}}{2}$

$4x^2 - 35 = q^2$

$(2x + 9)(2x - 9) = 35 = 7 \times 5$

$\Rightarrow x = 3$ or 9

n = 1, 9, 10 or 18

65. A

Sol. We have

$ar - 9 - (a - 5) = a(r - 1) - 4 = ar^2 - 25 - (ar - 9)$

$\Rightarrow \quad ar(r - 1) - 16$

We now have $ar^2 + (-2ar) + a = 12$

$\Rightarrow \quad 3ar = 432, ar = 144$

$\Rightarrow \quad r = \frac{4}{3} = 108.$

66. C

Sol. Let x be the probability that it rains on Sunday given that it does not rain on Saturday.

We then have $\frac{3}{5}x + \frac{2}{x}2x = \frac{3}{10} \Rightarrow x = \frac{3}{14}$

So probability that it does not rain on either day = $\left(1 - \frac{3}{14}\right) \cdot \frac{3}{5} = \frac{33}{70}$

So the probability that rains on at least one of the day is $1 - \frac{33}{70} = \frac{37}{70}$

67. B

Sol. All the coefficient of odd powered terms is an odd number of odd degree terms multiplied together and all coefficient of even powered terms has an even number of odd degree terms multiplied together. Since every odd degree term is negative and every even degree is positive so we sum is just $Q(-1) = P(-1)^5 = \frac{243}{32}$.

68. B

Sol. $xyz - 3 + \log_5 x = 32$ …(i)

$xyz - 3 + \log_5 y = 81$ …(ii)

$xyz - 3 + \log_5 z = 256$ …(iii)

Add (i), (ii) and (iii) we have

$xyz = 125$

$\Rightarrow \log_5 x = -90$

$\log_4 y = -41$

$\log_4 z = 134$

69. D

Sol. $\frac{ab+1}{a+b} < \frac{3}{2}$

$\Rightarrow 2ab + 2 < 3a + 3b$

$\Rightarrow (2a - 3)(2b - 3) < 5$

So $(2a - 3)$ and $(2b - 3)$ each cannot be even or else a and b will not be integer

$(2a - 3)(2b - 3) = (1, 1), (1, 3), (3, 1)$

$(a, b) = (2, 2), (2, 3), (3, 2)$

$\frac{a^3b^3+1}{a+b} = \frac{65}{16}$ or $\frac{31}{5}$. So Ans is $\frac{31}{5}$.

70. B

Sol. $x(y + z) = 0$

$\Rightarrow x = 0, y + z = 0$ pair of perpendicular plane.

71. D

Sol. Total number = $^7C_2 \cdot \frac{6!}{2!2!}$

Unit digit 3, $^6C_2 \cdot \frac{5!}{2!2!}$

Unit digit 1, $^6C_2 \cdot \frac{5!}{2!}$

Probability = $\frac{5}{14}$

72. D

Sol. $\alpha - \frac{1}{\beta\gamma} \Rightarrow \alpha\left(1 - \frac{1}{\alpha\beta\gamma}\right) = 2\alpha$

So, equation whose roots are 2α, 2β, 2γ are

$\left(\frac{x}{2}\right)^3 - 3\left(\frac{x}{2}\right) + 1 = 0$

$x^3 - 12x + 8 = 0$

73. B

Sol. $A_3 = 4$

$G_2 = \sqrt{5}$

$H_1 = \frac{5}{4}$

So equation is $(x - A_3)(x - G_2)(x - H_1) = 0$

74. B

Sol. Lines are coplanar if $\begin{vmatrix} 1 & 1 & 1 \\ 2 & x_1 & x_2 \\ 3 & 4 & 5 \end{vmatrix} = 0$

$\Rightarrow 2x_1 - x_2 = 2$

Also sum of roots = $\frac{-(x_2 + 2)}{x_1} = \frac{-2x_1}{x_1} = -2$.

75. D

Sol. Let $\vec{c} = \lambda\vec{a} + \mu\vec{b} + \nu(\vec{a} \times \vec{b})$, then

$\vec{a} \cdot \vec{c} = \lambda \Rightarrow \lambda = \cos\theta$.

Similarly $\mu = \cos\theta$.

also, $|\vec{c}|^2 = \lambda^2 + \mu^2 + \nu^2 |\vec{a} \times \vec{b}|^2$

$\Rightarrow 1 = 2\lambda^2 + \nu^2\left[|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2\right] = 2\lambda^2 + \nu^2$ $[\because \lambda = \mu]$

$\therefore \nu^2 = 1 - 2\lambda^2 = 1 - 2\cos^2\theta = -\cos 2\theta$.

Now $\nu^2 \geq 0 \Rightarrow \cos 2\theta \leq 0 \Rightarrow \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$.

76. A

Sol. $(1 - 2x)(1 - 2y)(1 - 2z) = 1 - 2\sum x + 4\sum xy - 8xyz$

$= 4\sum xy - 8xyz - 1$

$\therefore \sum xy - 2xyz = \frac{1 + (1-2x)(1-2y)(1-2z)}{4}$

$\frac{\sum(1-2x)}{3} \geq \{(1-2x)(1-2y)(1-2z)\}^{1/3}$.

$\Rightarrow (1 - 2x)(1 - 2y)(1 - 2z) \leq \frac{1}{27}$

$\Rightarrow xy + yz + zx - 2xyz \leq \frac{7}{27}$.

Equality holds for $x = y = z = \frac{1}{3}$

$\therefore$ maximum value = $\frac{7}{27}$.

77. B

Sol. $|A^{2005} - 6A^{2004}| = |A|^{2004} |A - 6I|$

$= 2^{2004} \cdot \begin{vmatrix} 0 & 11 \\ 2 & -2 \end{vmatrix} = (-22)\, 2^{2004} = (-)\, 2 \times 11 \times 2^{2004} = (-11)(2)^{2005}$.

78. B

Sol. $\Delta = \begin{vmatrix} 0 & -2b^2 & -2a^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$ $(R_1 \to R_1 - R_2 - R_3)$

$= -2\begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$

$= -2\begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & c^2 & 0 \\ b^2 & 0 & c^2 \end{vmatrix}$ $(R_1 \to R_2 - R_1 \text{ and } R_3 \to R_3 - R_1)$

$= 4a^2b^2c^2 = 4$ [as a, b, c are cube roots of unity].

79. B

Sol. First find the probability both are green, then the probability both are blue and add the two probability. The sum should be equal to 0.58.

$\Rightarrow \quad \frac{4}{10} \times \frac{16}{16+N} + \frac{6}{10} \cdot \frac{N}{16+N} = \frac{29}{50}$

$\Rightarrow \quad N = 144$

80. C

Sol. $(AB)^2 = AB.AB = A^3B^2$

$(AB)^3 = (AB)^2.AB = A^3B^2.AB = A^7.B^3$

$(AB)^4 = (AB)^3.AB = A^7.B^3.AB = A^{15}.B^4$

$\Rightarrow \quad (AB)^{10} = A^{1023}.B^{10}$

SECTION – B

81. 528

Sol. Let roots are $-x_1, -x_2, -x_3, -x_4$

$f(x) = (x + x_1)(x + x_2)(x + x_3)(x + x_4)$

$f(1) = (1 + x_1)(1 + x_2)(1 + x_3)(1 + x_4) = 1 + a + b + c + d = 2010$

Since $2010 = 2.3.5.6.7$

So d is the product of 4 roots $= (-1)(-2)(-4)(-66) = 528$

82. 9

Sol. Take

$x = 4 + 2\cos\theta$

$y = 3\sin\theta$

83. 24

Sol. Let $f(x) = x^4 q\left(\frac{1}{x}\right)$ it has same root as 1, 2, 3, 4 as p(x), but has same coefficient as q(x). Just in reverse order, it leading coefficient is $q(0) = \frac{1}{24}$, limit will be -24.

84. 2

Sol. On each roll, the probability that Kiran decides to go to Russia is $\frac{3}{2}$ times the probability she decides to go to Ukrain, so the total probability that she goes to Russia is $\frac{3}{2}$ times the totoal probability that she goes to Ukrain. Since the total probability is 1 (they are the only two eventual outcomes) Kiran goes to Ukrain with probability $\frac{2}{5}$ and Russia with probability $\frac{3}{5}$.

85. 8

Sol. Let $z = m + \sqrt{n}\,i$

$\overline{z} = m - \sqrt{n}\,i$

i.e. $x^3 + ax + b$ has roots -20, z, $\overline{z}$ and $x^3 + cx^2 + b$ has roots -21, z, $\overline{z}$

$\Rightarrow$ $-20 + z + \overline{z} = 0$

$m = 10$

$-21z - 21\overline{z} + z\overline{z} = 0$

$-21(z + \overline{z}) + z\overline{z} = 0$

$\Rightarrow$ $n = 320$

86. 121

Sol. $A.B = |A|.I$

$\Rightarrow$ $|A| = -11 \therefore |B| = |A|^2 = 121$

87. 3

Sol. The plane is $y - z = 0$

$a_1b_1 + a_2b_2 + a_3b_3 = 0 \Rightarrow k = -3$

88. 943

Sol. ${}^7P_4 \times 1 + \frac{{}^7P_3}{2} + 1 - 2 = 943$

89. 4

Sol. a, ar, ar^2 in G.P. $\Rightarrow ar^2 - ar < 1 \Rightarrow r^2 - r < 1 \Rightarrow r^2 - r - 1 < 0$

$r = \frac{\sqrt{5}+1}{2}$, $[r] = 1$, $[-r] = -2$

90. 6

Sol. Let $x - d, x, x + d$ one the sequence

So, it is given that $(x - d)^2 + x^2 + (x + d)^2 = xd^2 \Rightarrow 3x^2 + 2d^2 - xd^2 = 0 \Rightarrow 3x^2 - xd^2 + 2d^2 = 0$

$\Rightarrow x = \frac{d^2 \pm \sqrt{d^4 - 24d^2}}{6} = \frac{d^2 \pm d\sqrt{d^2 - 24}}{6}$

So, $d = 5$ and $d = 7$ are the only (+ve) integers value of d in possible, so that $\sqrt{d^2 - 24}$ a positive integer.

At $d = 5$, $x = 5$ and $x = \frac{10}{3}$, $d = 7$, $x = 14$ so, $(5 + 5) + (7 + 14) = 31$